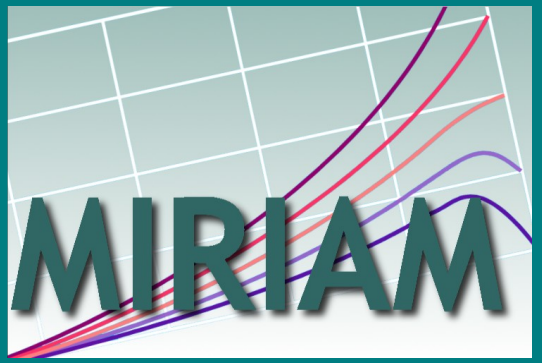
Identifiers.org and MIRIAM Registry: perennial identifiers for cross-referencing

Nick Juty

BioModels.net



EMBL-EBI

EBI is an Outstation of the European Molecular Biology Laboratory.

8th December 2011, SWAT4LS, London

Outline

General introduction of metadata

MIRIAM Registry of namespaces

- information captured
- supporting facilities
- export and access
- submissions

Identifiers.org resolving service

- levels of information
- information access



cross-references

Cross-references allow the **unambiguous identification** of:

- physical entities (records in a database)
  - molecular entities
  - diseases
  - clinical trials
  - ...
- information artefacts
  - ontological terms
  - scientific publications
  - books
    - ...
- ...



Use of cross-references

Cross-references, and generally **metadata**, are essential for:

understanding data

reusing data

comparing data

integrating data

converting data

providing efficient search strategies

...



Semantic Web: technologies

Set of technologies helping to link data of different natures and different locations in a meaningful way:

Uniform Resource Identifiers (URIs) to unambiguously identify pieces of data

Controlled vocabularies to characterise the relationships between data entity (SKOS, RDFs, ontologies)

Syntaxes to encode the **relationships** between data entity (RDF)

Query languages, to retrieve information encoded using semantic web technologies (SPARQL)



Towards globally unique identifiers

Namespace

Identifies a data collection

Entity identifier

Identifies a data entry within the data collection



Towards globally unique identifiers

Namespace

Identifies a data collection

from a shared list of *namespaces*

Entity identifier

Identifies a data entry within the data collection

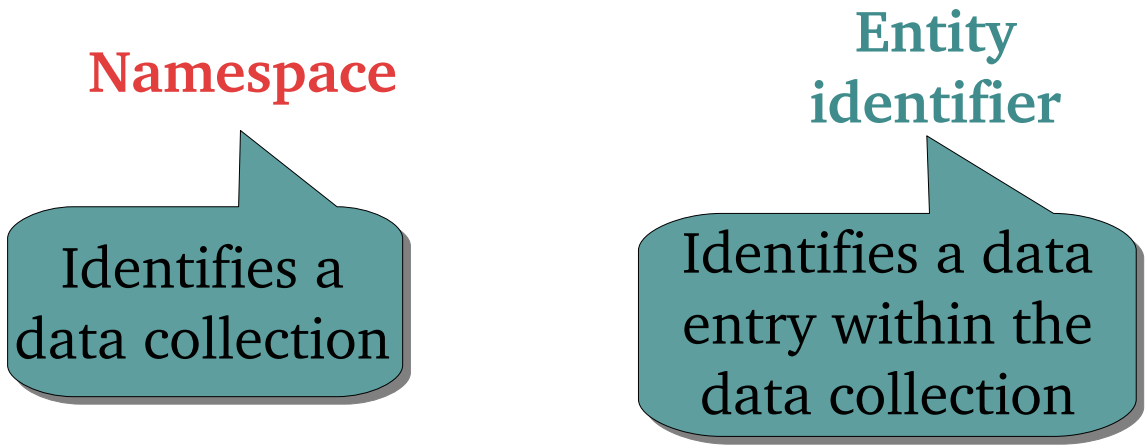
provided by the data collection

unique within the data collection

format defined by the data collection



Identifiers.org URLs



Human calmodulin: P62158 in UniProt

↳ `http://identifiers.org/uniprot/P62158`

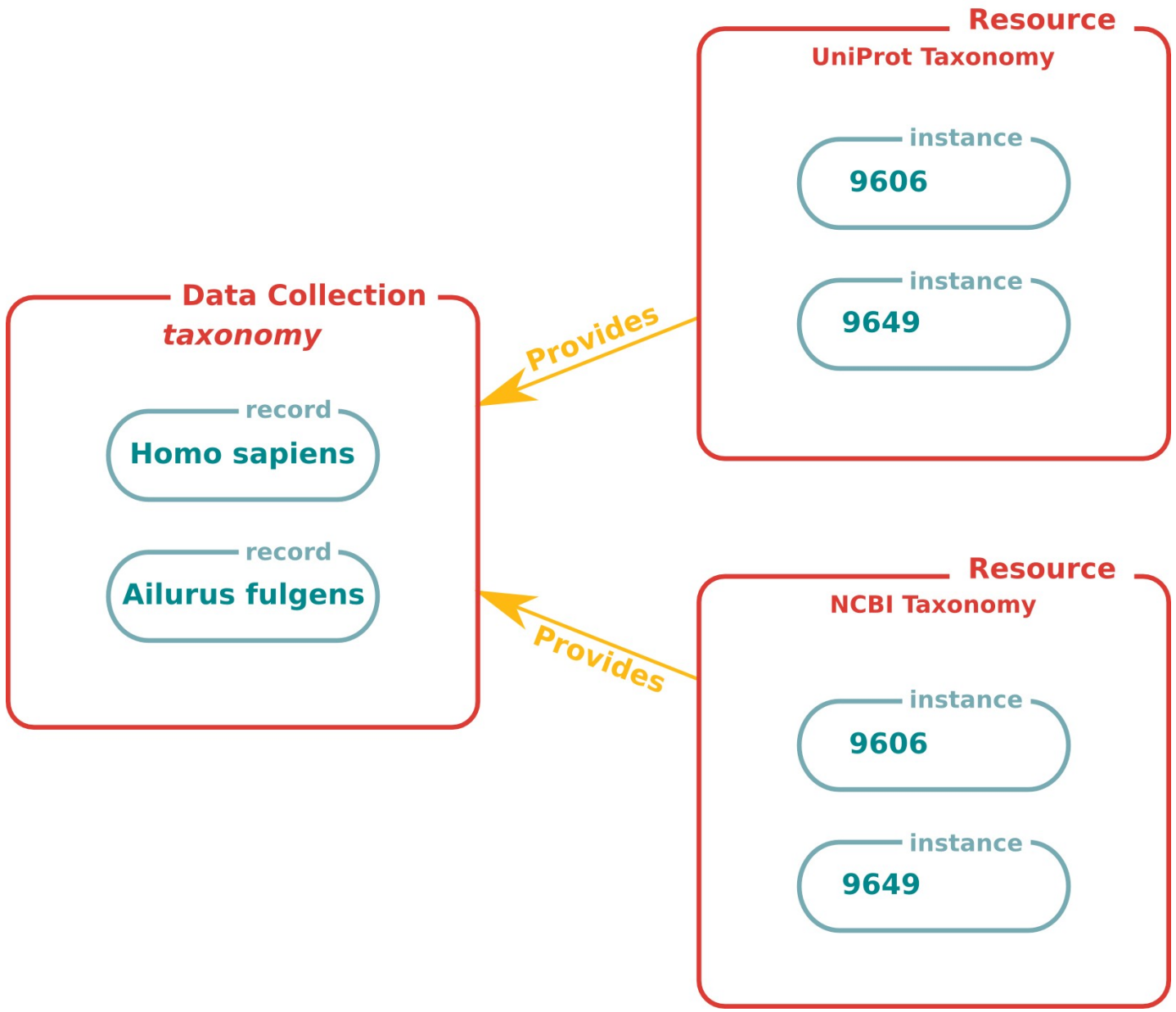
Alcohol dehydrogenase: 1.1.1.1 in EC code

↳ `http://identifiers.org/ec-code/1.1.1.1`

Activation of MAPKK activity: GO:0000186 in Gene Ontology

↳ `http://identifiers.org/obo.go/GO:0000186`

Nomenclature



MIRIAM Registry overview

- **MIRIAM Database**

  collection catalogue

  XML exports available

- **MIRIAM Web Services**

  programming interface for querying database

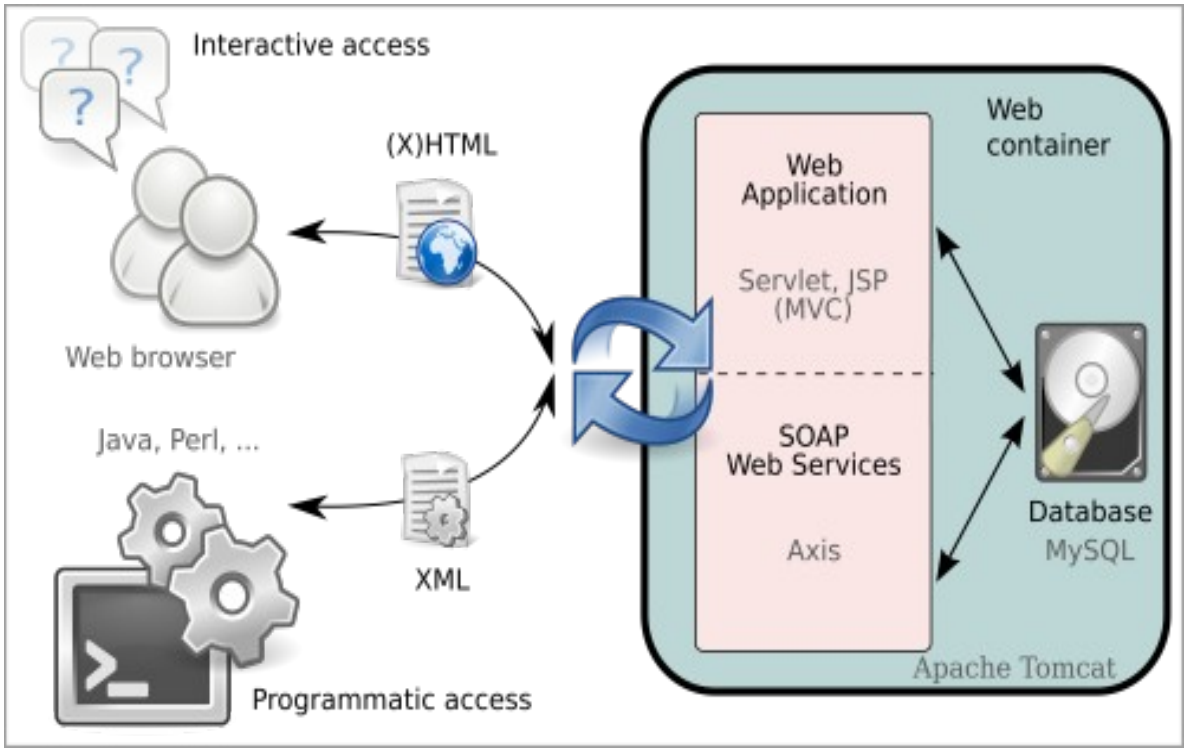
- **MIRIAM Library**

  ready to use Java code to query and access

- **MIRIAM Web Site**

  web page for browsing and querying

  allows new submission (curated)



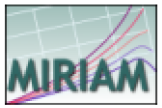
Laibe and Le Novère.
MIRIAM Resources: tools to generate and resolve robust cross-references in Systems Biology.
BMC Systems Biology, 2007

EMBL-EBI

MIRIAM Registry

EMBL-EBI | Enter Text Here | Find | Help | Feedback

Databases | Tools | Research | Training | Industry | About Us | Help | Site Index



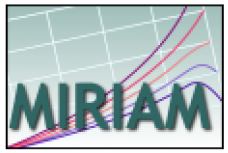
- Browse
- Search
- Tags
- Query services
- Submit new
- Export
- Curator Sign in

- Web Services
- Documents
  - MIRIAM Guidelines
  - FAQ
  - Documentation
  - Who's using MIRIAM?
  - Identification systems
  - News
  - BioModels.net Qualifiers

- MIRIAM on

EBI > Groups > Computational Neurobiology > Research > MIRIAM Registry

MIRIAM Registry



MIRIAM Registry are a set of online services created in support of MIRIAM, a set of guidelines for the annotation and curation of computational models.

The core of *MIRIAM Registry* is a catalogue of data types (namespaces corresponding to controlled vocabularies or databases), their URIs and the corresponding physical URLs or resources. Access to this data is made available via exports (XML) and Web Services (SOAP).

MIRIAM Registry is developed and maintained under the BioModels.net initiative, and are free for use by all.

Quick links

Browse

by data type name
by tags

Web Services

services available
usage of the services
online demonstration

Search

generic search

Exports

XML

http://www.ebi.ac.uk/miriam/



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List of data collections

Name	Namespace	Definition
Protein Model Database	pmdb	The Protein Model DataBase (PMDB), is a database that collects manually built three dimensional protein models, obtained by different structure prediction techniques.
MIRIAM Registry collection	miriam.collection	MIRIAM Registry is an online resource created to catalogue colelctions (Gene Ontology, Taxonomy or PubMed are some examples), their URIs and the corresponding resources (physical locations), whether these are controlled vocabularies or databases.
TEDDY	biomodels.teddy	The Terminology for Description of Dynamics (TEDDY) is an ontology for dynamical behaviours, observable dynamical phenomena, and control elements of bio-models and biological systems in Systems Biology and Synthetic Biology.
UniProt	uniprot	UniProt (Universal Protein Resource) is the world's most comprehensive catalog of information on proteins. It is a central repository of protein sequence and function created by joining the information contained in Swiss-Prot, TrEMBL, and PIR.
PubChem-substance	pubchem.substance	PubChem provides information on the biological activities of small molecules. It is a component of NIH's Molecular Libraries Roadmap Initiative. PubChem Substance archives chemical substance records.
EDAM Ontology	edam	EDAM is an ontology of general bioinformatics concepts, including topics, data types, formats, identifiers and operations. EDAM provides a controlled vocabulary for the description, in semantic terms, of things such as: web services (e.g. WSDL files), applications, tool collections and packages, work-benches and workflow software, databases and ontologies, XSD data schema and data objects, data syntax and file formats, web portals and pages, resource catalogues and documents (such as scientific publications).
Ontology for Biomedical Investigations	obo.obi	The Ontology for Biomedical Investigations (OBI) project is developing an integrated ontology for the description of biological and clinical investigations. The ontology will represent the design of an investigation, the protocols and instrumentation used, the material used, the data generated and the type analysis performed on it. Currently OBI is being built under the Basic Formal Ontology (BFO).
MACiE	macie	MACiE (Mechanism, Annotation and Classification in Enzymes) is a database of enzyme reaction mechanisms. Each entry in MACiE consists of an overall reaction describing the chemical compounds involved, as well as the species name in which the reaction occurs. The individual reaction stages for each overall reaction are listed with mechanisms, alternative mechanisms, and amino acids involved.
FMA	obo.fma	The Foundational Model of Anatomy Ontology (FMA) is a biomedical informatics ontology. It is concerned with the representation of classes or types and relationships necessary for the symbolic representation of the

EMBL-EBI

… and their associated *namespace*

Name	Namespace	Definition
Protein Model Database	pmdb	The Protein Model DataBase (PMDB), is a database that collects manually built three dimensional protein models, obtained by different structure prediction techniques.
MIRIAM Registry collection	miriam.collection	MIRIAM Registry is an online resource created to catalogue colelctions (Gene Ontology, Taxonomy or PubMed are some examples), their URIs and the corresponding resources (physical locations), whether these are controlled vocabularies or databases.
TEDDY	biomodels.teddy	The Terminology for Description of Dynamics (TEDDY) is an ontology for dynamical behaviours, observable dynamical phenomena, and control elements of bio-models and biological systems in Systems Biology and Synthetic Biology.
UniProt	uniprot	UniProt (Universal Protein Resource) is the world's most comprehensive catalog of information on proteins. It is a central repository of protein sequence and function created by joining the information contained in Swiss-Prot, TrEMBL, and PIR.
PubChem-substance	pubchem.substance	PubChem provides information on the biological activities of small molecules. It is a component of NIH's Molecular Libraries Roadmap Initiative. PubChem Substance archives chemical substance records.
EDAM Ontology	edam	EDAM is an ontology of general bioinformatics concepts, including topics, data types, formats, identifiers and operations. EDAM provides a controlled vocabulary for the description, in semantic terms, of things such as: web services (e.g. WSDL files), applications, tool collections and packages, work-benches and workflow software, databases and ontologies, XSD data schema and data objects, data syntax and file formats, web portals and pages, resource catalogues and documents (such as scientific publications).
Ontology for Biomedical Investigations	obo.obi	The Ontology for Biomedical Investigations (OBI) project is developing an integrated ontology for the description of biological and clinical investigations. The ontology will represent the design of an investigation, the protocols and instrumentation used, the material used, the data generated and the type analysis performed on it. Currently OBI is being built under the Basic Formal Ontology (BFO).
MACiE	macie	MACiE (Mechanism, Annotation and Classification in Enzymes) is a database of enzyme reaction mechanisms. Each entry in MACiE consists of an overall reaction describing the chemical compounds involved, as well as the species name in which the reaction occurs. The individual reaction stages for each overall reaction are listed with mechanisms, alternative mechanisms, and amino acids involved.
FMA	obo.fma	The Foundational Model of Anatomy Ontology (FMA) is a biomedical informatics ontology. It is concerned with the representation of classes or types and relationships necessary for the symbolic representation of the

Data collection: *Enzyme Nomenclature*

Overview | Categories | Miscellaneous

Overview of the data collection

Name		
Identifier		MIR:00000004
Name		Enzyme Nomenclature
Synonyms		Enzyme Classification
		EC code
		EC
Information		
Definition		The Enzyme Classification contains the recommendations of the Nomenclature Committee of the International Union of Biochemistry and Molecular Biology on the nomenclature and classification of enzyme-catalysed reactions.
Identifier pattern		^\d+\.-\.-\.-|\d+\.\d+\.-\.-|\d+\.\d+\.\d+\.-|\d+\.\d+\.\d+\.(n)?\d+$
URIs		
Namespace		ec-code
Root URL ⊞		http://identifiers.org/ec-code/
Root URN ⊞		urn:miriam:ec-code:
Physical Locations		
Resource MIR:00100002	**Access URL**	http://www.genome.jp/dbget-bin/www_bget?ec:**$id** [Example: 1.1.1.1]
	Website	http://www.genome.jp/dbget-bin/www_bfind?enzyme
	Description	KEGG Ligand Database for Enzyme Nomenclature
	Institution	Kyoto University Bioinformatics Center, Japan
Resource MIR:00100003	**Access URL**	http://enzyme.expasy.org/EC/**$id** [Example: 1.1.1.1]
	Website	http://enzyme.expasy.org/
	Description	Enzyme nomenclature database, ExPASy (Expert Protein Analysis System)
	Institution	Swiss Institute of Bioinformatics, Switzerland
Resource MIR:00100001	**Access URL**	http://www.ebi.ac.uk/intenz/query?cmd=SearchEC&ec=**$id** [Example: 1.1.1.1]
	Website	http://www.ebi.ac.uk/intenz/
	Description	IntEnZ (Integrated relational Enzyme database)
	Institution	European Bioinformatics Institute, United Kingdom
References		
http://www.chem.qmul.ac.uk/iubmb/enzyme/		
⊞ http://srs.ebi.ac.uk/srsbin/cgi-bin/wgetz?-view+MedlineFull+[medline-PMID:10812475]		

Data collection: *Enzyme Nomenclature*

Overview | Categories | Miscellaneous

Overview of the data collection

Name	
Identifier	MIR:00000004
Name	Enzyme Nomenclature
Synonyms	Enzyme Classification
	EC code
	EC
Information	
Definition	The Enzyme Classification contains the recommendations of the Nomenclature Committee of the International Union of Biochemistry and Molecular Biology on the nomenclature and classification of enzyme-catalysed reactions.
Identifier pattern	^\d+\.-\.-\.-|\d+\.\d+\.-\.-|\d+\.\d+\.\d+\.-|\d+\.\d+\.\d+\.(n)?\d+$
URIs	
Namespace	ec-code
Root URL ⊞	http://identifiers.org/ec-code/
Root URN ⊞	urn:miriam:ec-code:
Physical Locations	
Resource MIR:00100002 — **Access URL**	http://www.genome.jp/dbget-bin/www_bget?ec:**$id** [Example: 1.1.1.1]
Website	http://www.genome.jp/dbget-bin/www_bfind?enzyme
Description	KEGG Ligand Database for Enzyme Nomenclature
Institution	Kyoto University Bioinformatics Center, Japan
Resource MIR:00100003 — **Access URL**	http://enzyme.expasy.org/EC/**$id** [Example: 1.1.1.1]
Website	http://enzyme.expasy.org/
Description	Enzyme nomenclature database, ExPASy (Expert Protein Analysis System)
Institution	Swiss Institute of Bioinformatics, Switzerland
Resource MIR:00100001 — **Access URL**	http://www.ebi.ac.uk/intenz/query?cmd=SearchEC&ec=**$id** [Example: 1.1.1.1]
Website	http://www.ebi.ac.uk/intenz/
Description	IntEnZ (Integrated relational Enzyme database)
Institution	European Bioinformatics Institute, United Kingdom
References	
http://www.chem.qmul.ac.uk/iubmb/enzyme/	
⊞ http://srs.ebi.ac.uk/srsbin/cgi-bin/wgetz?-view+MedlineFull+[medline-PMID:10812475]	

Resources

… and associated resources

		Physical Locations
Resource MIR:00100002	**Access URL**	http://www.genome.jp/dbget-bin/www_bget?ec:**$id** [Example: 1.1.1.1]
	Website	http://www.genome.jp/dbget-bin/www_bfind?enzyme
	Description	KEGG Ligand Database for Enzyme Nomenclature
	Institution	Kyoto University Bioinformatics Center, Japan
Resource MIR:00100003	**Access URL**	http://enzyme.expasy.org/EC/**$id** [Example: 1.1.1.1]
	Website	http://enzyme.expasy.org/
	Description	Enzyme nomenclature database, ExPASy (Expert Protein Analysis System)
	Institution	Swiss Institute of Bioinformatics, Switzerland
Resource MIR:00100001	**Access URL**	http://www.ebi.ac.uk/intenz/query?cmd=SearchEC&ec=**$id** [Example: 1.1.1.1]
	Website	http://www.ebi.ac.uk/intenz/
	Description	IntEnZ (Integrated relational Enzyme database)
	Institution	European Bioinformatics Institute, United Kingdom

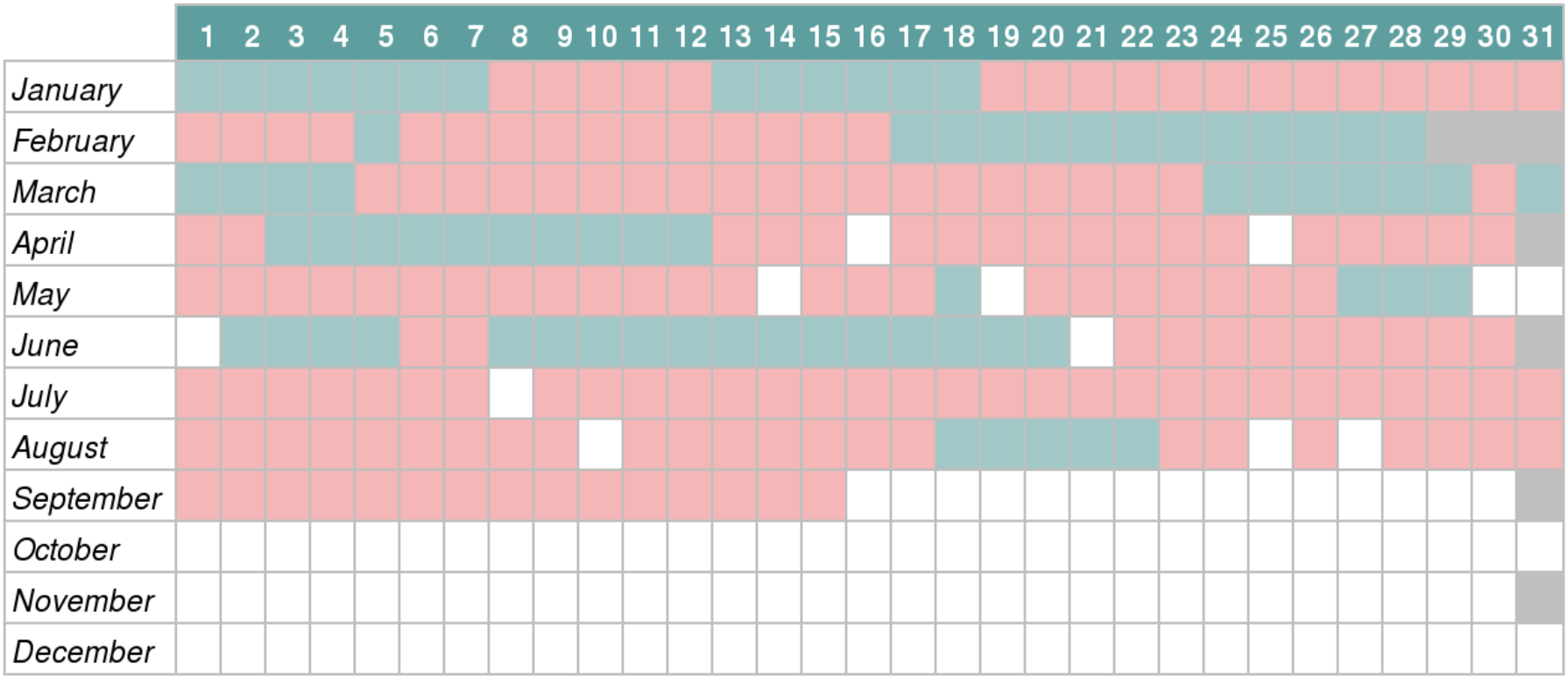


Resource reliability

Health history: MIR:[redacted]

Health history of: MIR:[redacted]

2010



2009

Resource: MIR:00100050

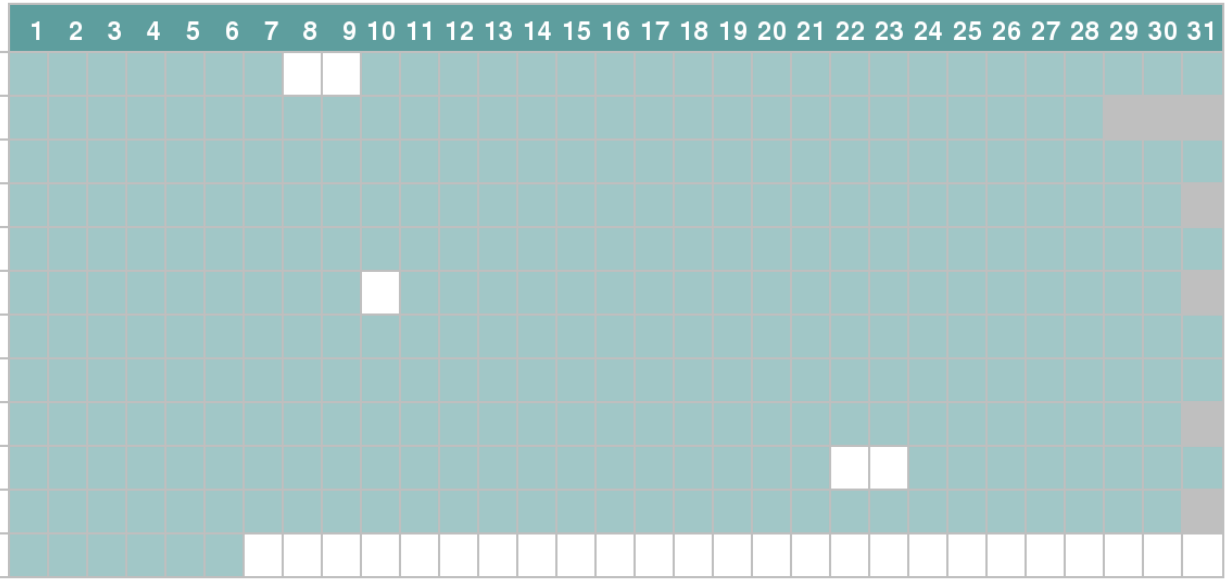
General information about the resource: ***The FlyBase Database*** (associated with the data collection: FlyBase).

	Health statistics
Last known state	up
Last check	2011-12-06 06:33:22
Uptime ratio	100% (1012 checks)
Downtime ratio	0% (0 checks)
Unknown ratio	0% (0 checks)
URL used	http://www.flybase.org/reports/FBgn0011293.html

Health history

Full record of the health checks performed on this resource.

2011



Submit a new data collection

Please fill this form in order to submit a new data type to MIRIAM Registry. Alternatively, you can contact us with your query.

You **do not** need to fill all the fields, you can just enter the information you have.

The new data type will not be directly publicly available after you pressed the **Submit** button. A curator will first check if it complies with the terms and conditions of MIRIAM Registry and, if necessary, correct and complete it before publishing it.

Help

You can display all help bubbles by clicking on: Displays all the help messages, or hide them: Hides all the help messages.

Moreover, you can display the individual help by clicking on the button: located in the title of each section.

Information about the new data type

First you need to enter the name of the data type you want to add to the database. After you can add as much synonyms as you want.

Name and synonyms

Primary name:

[Add a synonym]

Here is some information about the data type: definition and regular expression (*i.e.* pattern for identifiers of elements, following the PERL style).

Definition and pattern

Definition: Enter definition here...

Identifier pattern: Enter Identifier pattern here...

Submit a new data collection

Please fill this form in order to submit a new data type to MIRIAM Registry. Alternatively, you can contact us with your query.

You **do not** need to fill all the fields, you can just enter the information you have.

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Name and synonyms

Primary name:

[Add a synonym]

Here is some information about the data type: definition and regular expression (*i.e.* pattern for identifiers of elements, following the PERL style).

Definition and pattern

Definition: Enter definition here...

Identifier pattern: Enter Identifier pattern here...

Registry: suggest changes

Overview | Categories | Miscellaneous

Overview of the data collection

Name		
Identifier		MIR:00000004
Name		Enzyme Nomenclature
Synonyms		Enzyme Classification
		EC code
		EC
Information		
Definition		The Enzyme Classification contains the recommendations of the Nomenclature Committee of the International Union of Biochemistry and Molecular Biology on the nomenclature and classification of enzyme-catalysed reactions.
Identifier pattern		
Namespace		
Root URL		
Root URN		urn:miriam:ec-code:
Physical Locations		
Resource MIR:00100002	**Access URL**	http://www.genome.jp/dbget-bin/www_bget?ec:**$id** [Example: 1.1.1.1]
	Website	http://www.genome.jp/dbget-bin/www_bfind?enzyme
	Description	KEGG Ligand Database for Enzyme Nomenclature
	Institution	Kyoto University Bioinformatics Center, Japan
Resource MIR:00100003	**Access URL**	http://enzyme.expasy.org/EC/**$id** [Example: 1.1.1.1]
	Website	http://enzyme.expasy.org/
	Description	Enzyme nomenclature database, ExPASy (Expert Protein Analysis System)
	Institution	Swiss Institute of Bioinformatics, Switzerland
Resource MIR:00100001	**Access URL**	http://www.ebi.ac.uk/intenz/query?cmd=SearchEC&ec=**$id** [Example: 1.1.1.1]
	Website	http://www.ebi.ac.uk/intenz/
	Description	IntEnZ (Integrated relational Enzyme database)
	Institution	European Bioinformatics Institute, United Kingdom
References		
http://www.chem.qmul.ac.uk/iubmb/enzyme/		
⊞ http://srs.ebi.ac.uk/srsbin/cgi-bin/wgetz?-view+MedlineFull+[medline-PMID:10812475]		

Suggest modifications to this data collection

Go back to the list of data collections

Suggest modifications to this data collection

Registry: summary

list of **data collections**

web services and **XML export** available

community driven (everybody can submit new data collections or suggest modification of existing records)

curated resource

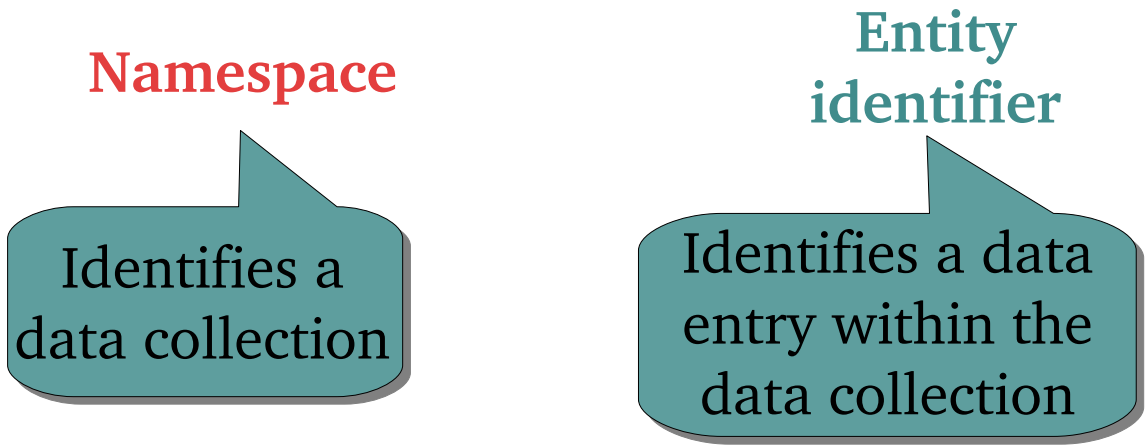
systems in place to **monitor** registered web resources

unrestricted scope (currently mainly focused on Life Sciences, but the scope is potentially unlimited)

free to use



Identifiers.org URLs



Human calmodulin: P62158 in UniProt

↳ `http://identifiers.org/uniprot/P62158`

Alcohol dehydrogenase: 1.1.1.1 in EC code

↳ `http://identifiers.org/ec-code/1.1.1.1`

Activation of MAPKK activity: GO:0000186 in Gene Ontology

↳ `http://identifiers.org/obo.go/GO:0000186`

identifiers.org

Home News Help Examples Registry About

Welcome to Identifiers.org!

Identifiers.org is a system providing resolvable persistent URIs used to identify data for the scientific community, with a current focus on the Life Sciences domain. The provision of a resolvable identifiers (URLs) fits well with the Semantic Web vision, and the Linked Data initiative.

Links

http://identifiers.org/

- About
- Help
- Examples URIs
- MIRIAM Registry

Board of trustees

Identifiers.org is a community project which activities are overseen by the following board of trustees:

- Michel Dumontier (Carleton University, Ottawa, Canada - Bio2RDF, W3C HCLS)
- Pascale Gaudet (Swiss Institute of Bioinformatics, Geneva, Switzerland - BioDBCore)
- Michael Hucka (California Institute of Technology, Pasadena, USA - SBML)
- Nicolas Le Novère (EMBL-EBI, Hinxton, UK - BioModels Database)
- Philippe Rocca-Serra (Oxford University, Oxford, UK - BioSharing)
- Mark Wilkinson (St. Paul's Hospital/UBC Vancouver, Canada - LSRN, SADI)

Contact

If you have any queries or experience any issues with this service, please contact: **biomodels-net-support [AT] lists.sf.net**



Identifiers.org: basic identification

Basic access

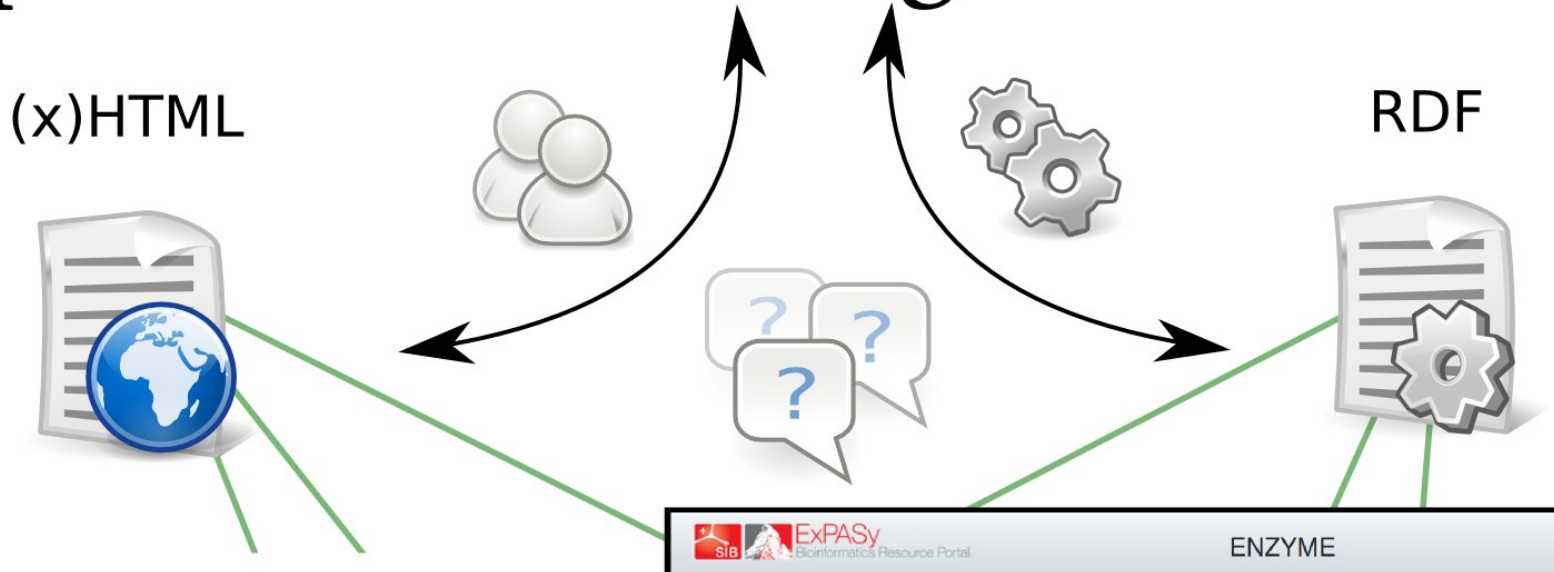
Collection:
http://identifiers.org/ec-code/

Record (location independent):
http://identifiers.org/ec-code/1.1.1.1



Identifiers.org: resolving

http://identifiers.org/ec-code/1.1.1.1



KEGG

ENZYME: 1.1.1.1

Entry	EC 1.1.1.1 Enzyme
Name	alcohol dehydrogenase; aldehyde reductase; ADH; alcohol dehydrogenase (NAD); aliphatic alcohol dehydrogenase; ethanol dehydrogenase; NAD-dependent alcohol dehydrogenase; NAD-specific aroma NADH-alcohol dehyd NADH-aldehyde dehy primary alcohol de yeast alcohol dehy
Class	Oxidoreductases; Acting on the CH-O With NAD+ or NADP+ BRITE hierarchy
Sysname	alcohol:NAD+ oxido
Reaction(IUBMB)	(1) a primary alco (2) a secondary al
Reaction(KEGG)	R07326 > R00623 R0 R06927 R08281 R083

ExPASy Bioinformatics Resource Portal — ENZYME — Home | Contact

ENZYME entry: EC 1.1.1.1

Accepted Name

Alcohol dehydrogenase.

Alternative Name(s)

Aldehyde reductase.

Reaction catalysed

An alcohol + NAD(+) <=> an aldehyde or ketone + NADH

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IntEnz

Quick search: Search

- IntEnz home
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- Data submission
- Downloads
- Documentation

EBI > Databases > Enzymes > IntEnz

EC 1 - Oxidoreductases

EC 1.1 - Acting on the CH-OH Group of Donors

EC 1.1.1 - With NAD+ or NADP+ as acceptor

EC 1.1.1.1 - Alcohol dehydrogenase

IntEnz view | ENZYME view

XML

IntEnz Enzyme Nomenclature

EC 1.1.1.1

EMBL-EBI

Choice of resources (HTML version)

http://identifiers.org/ec-code/1.1.1.1

3 physical locations (or resources) are available for accessing *1.1.1.1* (from Enzyme Nomenclature):

IntEnZ (Integrated relational Enzyme database)
European Bioinformatics Institute

United Kingdom

(Uptime: 100%)

Enzyme nomenclature database, ExPASy (Expert Protein Analysis System)
Swiss Institute of Bioinformatics

Switzerland

(Uptime: 99%)

KEGG Ligand Database for Enzyme Nomenclature
Kyoto University Bioinformatics Center

Japan

(Uptime: 100%)

Identifiers.org: more identification

Custom queries + format:

Response format:

http://identifiers.org/ec-code/1.1.1.1?format=rdfxml

Specified resource:

http://identifiers.org/ec-code/1.1.1.1?resource=MIR:00100001

profiles (predefined resolution locations):

http://identifiers.org/ec-code/1.1.1.1?profile=demo



http://identifiers.org/pubmed/16333295

http://identifiers.org/pubmed/16333295

4 physical locations (or resources) are available for accessing *16333295* (from PubMed):

HubMed
Alfred D. Eaton

United Kingdom

(Uptime: 97%)

SRS@EBI
European Bioinformatics Institute

United Kingdom

(Uptime: 100%)

CiteXplore
European Bioinformatics Institute

United Kingdom

(Uptime: 99%)

free digital archive of biomedical and life sciences journal literature
National Center for Biotechnology Information

USA

(Uptime: 100%)

Resolving: using a specific resource

http://identifiers.org/pubmed/16333295?resource=MIR:00100032

http://identifiers.org/pubmed/16333295 Close

Access to 16333295 (from PubMed) using the resource MIR:00100032.
Entity available from **4** providers, for more information please refer to: http://identifiers.org/pubmed/16333295.

Powered by: Identifiers.org & MIRIAM Registry

CiteXplore

- Quick search
- Advanced Search
- Query History
- Web services
- Help
- Statistics
- Medline
- C.B.A.
- E.P.O.
- Citeseer
- Whatizit
- iHOP

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Pubmed Id	16333295
Title	**Minimum information requested in the annotation of biochemical models (MIRIAM).**
Authors	Le Novère N., Finney A., Hucka M., Bhalla US., Campagne F., Collado-Vides J., Crampin EJ., Halstead M., Klipp E., Mendes P. See all ..
Affiliation	European Bioinformatics Institute, Hinxton, CB10 1SD, UK. lenov@ebi.ac.uk
Language	English
Journal	Nat. Biotechnol. (ISSN:1087-0156) (ESSN: 1546-1696) [2005 Dec; Volume: 23 (Issue: 12)] Page info: 1509-15
Publication type	Journal Article;
Full text article	doi>
XML	XML

Abstract

Most of the published quantitative models in biology are lost for the community because they are either not made available or they are insufficiently characterized to allow them to be reused. The lack of a standard description format, lack of stringent reviewing and authors' carelessness are the main causes for incomplete model descriptions. With today's increased interest in detailed biochemical models, it is necessary to define a minimum quality standard for the encoding of those models. We propose a set of rules for curating quantitative models of biological systems. These rules define procedures for encoding and annotating models represented in machine-readable form. We believe their application will enable users to (i) have confidence that curated models are an accurate reflection of their associated reference descriptions, (ii) search collections of curated models with precision, (iii) quickly identify the biological phenomena that a given curated model or model constituent represents and (iv) facilitate model reuse and composition into large subcellular models.

Cited by

Cited: 102 Times | Web of Science? Times Cited:146

Dry work in a wet world: computation in systems biology. 4 cite(s)

Kahlem P, Birney E. *Mol. Syst. Biol. (2006)*

Show Context

Profiles:

list of data collections

subset of the Registry's content

each data collection in a profile can have settings (such as **one** preferred resource)

can be defined at the level of an institution, project, individual, ...

can be public (shareable) or private (protected by a key)

centrally managed in the Registry (*user interface in progress*)

have a unique shortname (used for identification purposes in URLs)

provide access to a custom XML export

Provide access to custom web services (*in progress*)



http://identifiers.org/obo.go/GO:0006915

http://identifiers.org/obo.go/GO:0006915

4 physical locations (or resources) are available for accessing *GO:0006915* (from Gene Ontology):

GO Browser
The Jackson Laboratory

USA

(Uptime: 100%)

QuickGO (Gene Ontology browser)
European Bioinformatics Institute

United Kingdom

(Uptime: 100%)

GO through BioPortal
National Center for Biomedical Ontology, Stanford

USA

(Uptime: 97%)

AmiGO
The Gene Ontology Consortium

USA

(Uptime: 99%)

Resolving using a specific profile

http://identifiers.org/obo.go/GO:0006915?profile=demo

http://identifiers.org/obo.go/GO:0006915

Close

Access to GO:0006915 (from Gene Ontology) using the preferred resource of the profile demo.
Entity available from **4** providers, for more information please refer to: http://identifiers.org/obo.go/GO:0006915.

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GO:0006915 apoptosis

QuickGO | Click for example search | Search! | Web Services | Dataset | Term Basket: 0

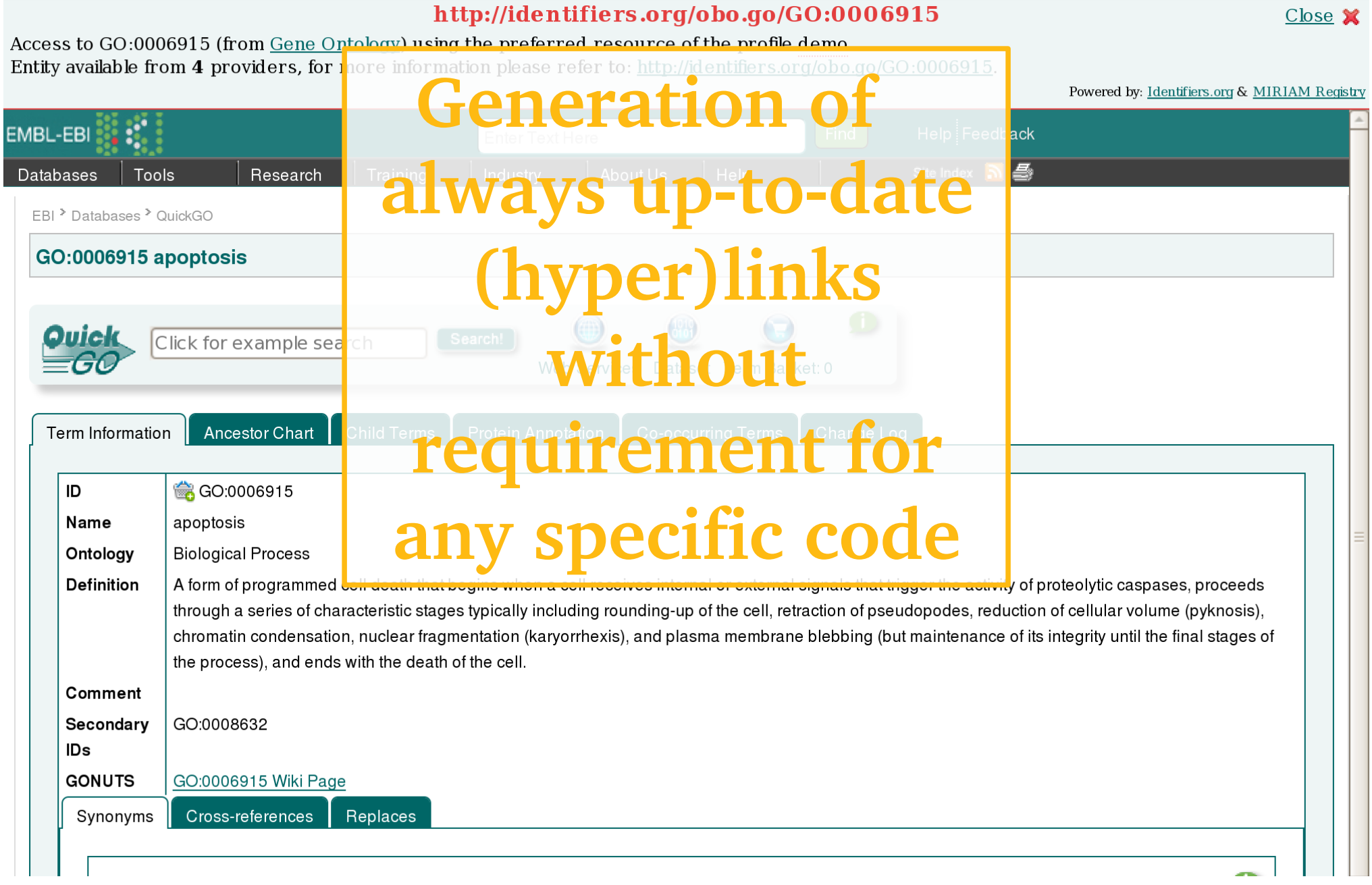
Term Information | Ancestor Chart | Child Terms | Protein Annotation | Co-occurring Terms | Change Log

ID	GO:0006915
Name	apoptosis
Ontology	Biological Process
Definition	A form of programmed cell death that begins when a cell receives internal or external signals that trigger the activity of proteolytic caspases, proceeds through a series of characteristic stages typically including rounding-up of the cell, retraction of pseudopodes, reduction of cellular volume (pyknosis), chromatin condensation, nuclear fragmentation (karyorrhexis), and plasma membrane blebbing (but maintenance of its integrity until the final stages of the process), and ends with the death of the cell.
Comment	
Secondary IDs	GO:0008632
GONUTS	GO:0006915 Wiki Page

Synonyms | Cross-references | Replaces

Resolving using a specific profile

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Generation of always up-to-date (hyper)links without requirement for any specific code

Resolving: using the most reliable resource

http://identifiers.org/ec-code/1.1.1.1?profile=most_reliable

http://identifiers.org/ec-code/1.1.1.1 Close

Access to 1.1.1.1 (from Enzyme Nomenclature) using the preferred resource of the profile most_reliable.
Entity available from **3** providers, for more information please refer to: http://identifiers.org/ec-code/1.1.1.1.

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EC 1 - Oxidoreductases

EC 1.1 - Acting on the CH-OH Group of Donors

EC 1.1.1 - With NAD+ or NADP+ as acceptor

EC 1.1.1.1 - Alcohol dehydrogenase

IntEnz view | ENZYME view

XML

IntEnz Enzyme Nomenclature

EC 1.1.1.1

Names

Accepted name: alcohol dehydrogenase

Other names: ADH
NAD-dependent alcohol dehydrogenase
NAD-specific aromatic alcohol dehydrogenase
NADH-alcohol dehydrogenase
NADH-aldehyde dehydrogenase
alcohol dehydrogenase (NAD)
aldehyde reductase
aliphatic alcohol dehydrogenase
ethanol dehydrogenase

Identifiers.org: short summary

provides **uniform identifiers** for Life Sciences

- **unambiguous**
- **perennial**
- **directly resolvable**
- standard compliant (**URIs**)
- location independent

identifiers available at **multiple levels** (data collection, resource, and data record)

customisable behaviours (formats available, preferred resource, ...)

responses encoded in **HTML** and **RDF** (either requested explicitly in the URLs or via *content negotiation*)

built on the ***MIRIAM Registry***



What to store in the database?

http://identifiers.org/ec-code/1.1.1.1



What to use in your import/export formats?

http://identifiers.org/ec-code/1.1.1.1



What to use in your user interfaces?

http://identifiers.org/ec-code/1.1.1.1



Where to get the data?

http://identifiers.org/ec-code/1.1.1.1



Acknowledgements

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Identifiers.org board of trustees

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